

Algorithmic

Post 3 months ago - Stop drawing lines and start reading the tape. How I use Delta Divergence & Order Flow to catch reversals (No Astrology)

Look, I'm not here to bash anyone's setup. If you're making money drawing triangles and waiting for the RSI to cross, good for you. Sseriously. But for me (I don't think it's just me), that stuff was pure astrology. I spent two years getting chopped up trying to trade "head and shoulders" patterns that the algos just ripped right through. It was hilarious and psinful to watch.

Started printing when I stopped trying to predict where price **should** go and started looking at what the participants were **actually doing**.

I want to break down the only setup I really trade now (it's solid, haven't changed a thing in months). It's a mean reversion play based on **order flow** and **delta divergence** (yes, the good ol' delta div). No lagging indicators, no magic fib spirals or whatever. Just buyers vs. sellers

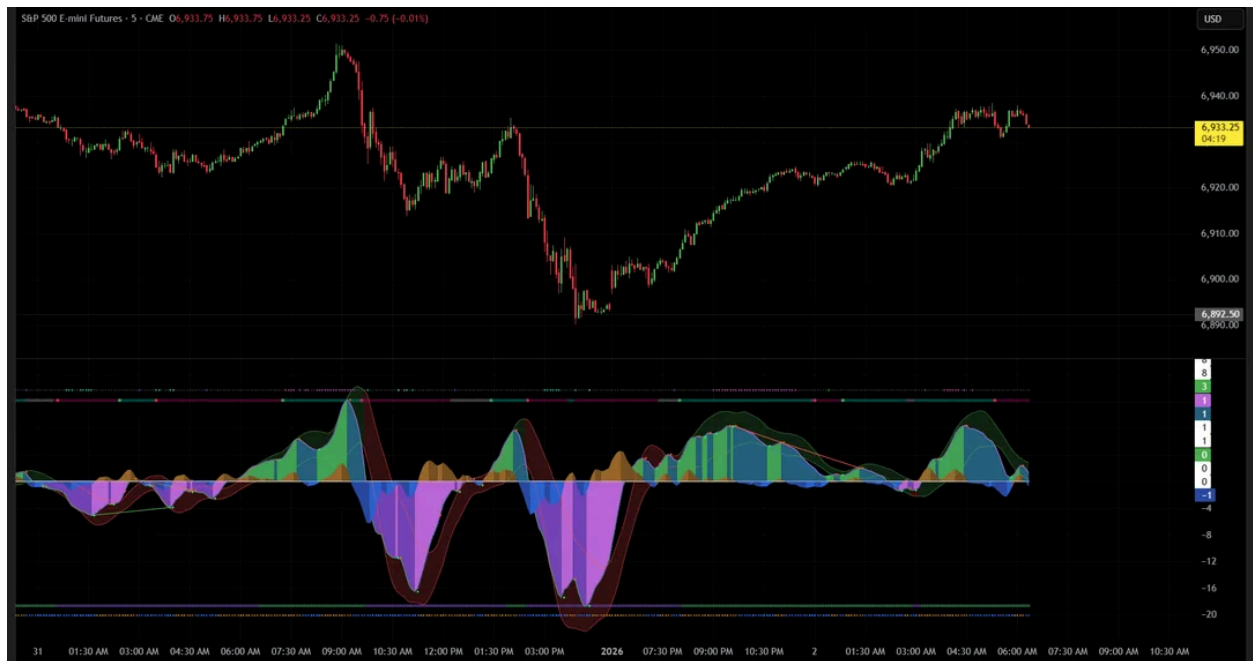
The Logic (Auction Market Theory)

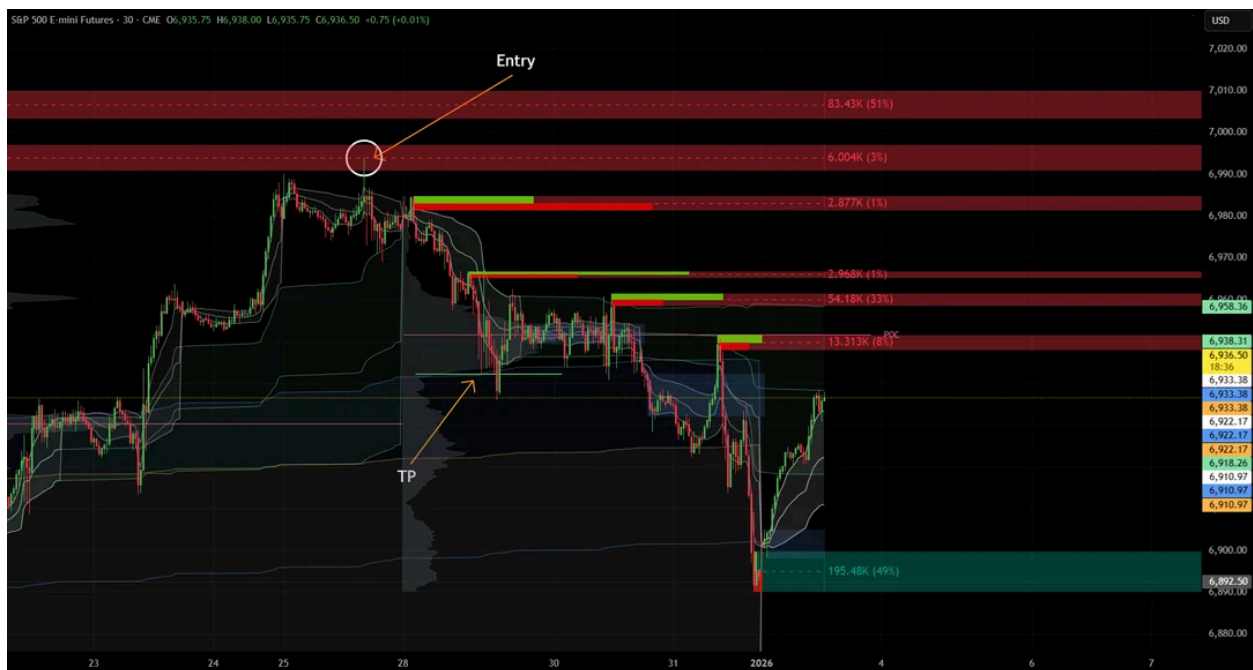
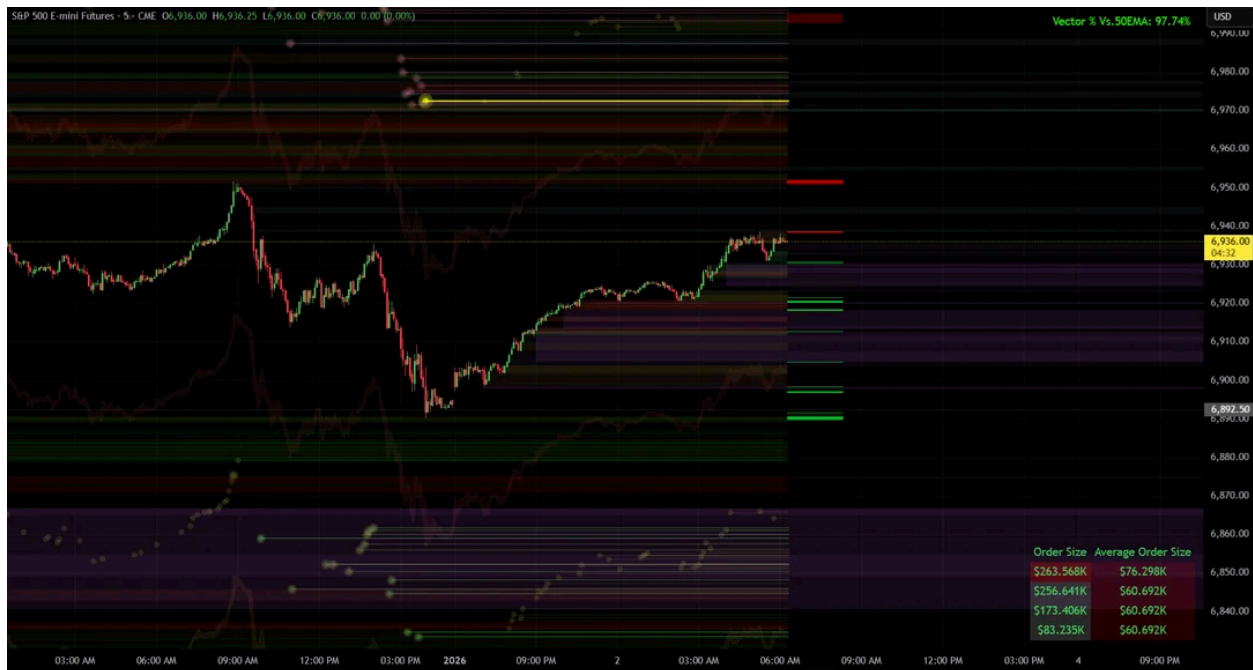
In shortest and simplest possible terms, price moves to advertise for business. When it goes up, it's looking for sellers. When it goes down, it's looking for buyers.

These are absolute basics, yet I notice that most traders don't pay attention to this.

Most people try to guess the bottom by drawing a support line. The problem is, you don't know if that support is made out of concrete or paper until it breaks.

Instead, I look for **Absorption** (!).





The Setup: Delta Divergence

Check the screenshots I attached. This is what I'm looking for:

1. **Price makes a new low:** The chart looks ugly. candles are red. Retail is clearly panic selling or getting stopped out

2. **Delta makes a higher low (or goes positive unproportional to the price change):** This is the key, because "Delta" is just aggressive buys minus aggressive sells.
3. **The Signal:** If price is dropping, but the delta is rising (check the bottom histogram), it means passive buyers (limit orders) are absorbing all that aggressive selling. The sellers dumping all the liquidity they have, but the price won't go lower.

That's the trap. Once the sellers run out of ammo, the auction has to rotate up to find liquidity that makes it function. Basic physics.

The "semi-auto" approach

I realized pretty quickly that my eyes are too slow to catch this in real-time because it moves like a caffeinated prison escapee (horrible comparison)

I built a semi-automated system (basically a custom footprint/heatmap overlay that also generates signals and sends alerts/notifications) that highlights these imbalances for me. It tracks the delta divergence and spots when the order book (heatmap (it's an est)) is stacking liquidity on one side.

It doesn't trade for (me I don't trust full auto bots) but it flags the zone. It basically screams ABSORPTION so I don't have to stare at the tape all day. It's the difference between guessing a reversal with TA / subjective patterns and seeing the order flow confirm it.

Why I'm posting this

I see a lot of people here asking "why did my support level fail?"

The answer is usually in the order flow. The level failed because there were no passive buyers there to soak up the hits.

It's infuriating to see so many people asking it and sometimes not even getting the answer.

If you're tired of the astrology stuff, look into order flow, cumulative delta (CVD), and footprint charts. It's a steep learning curve, but it's the only way to see the "why" behind the move.

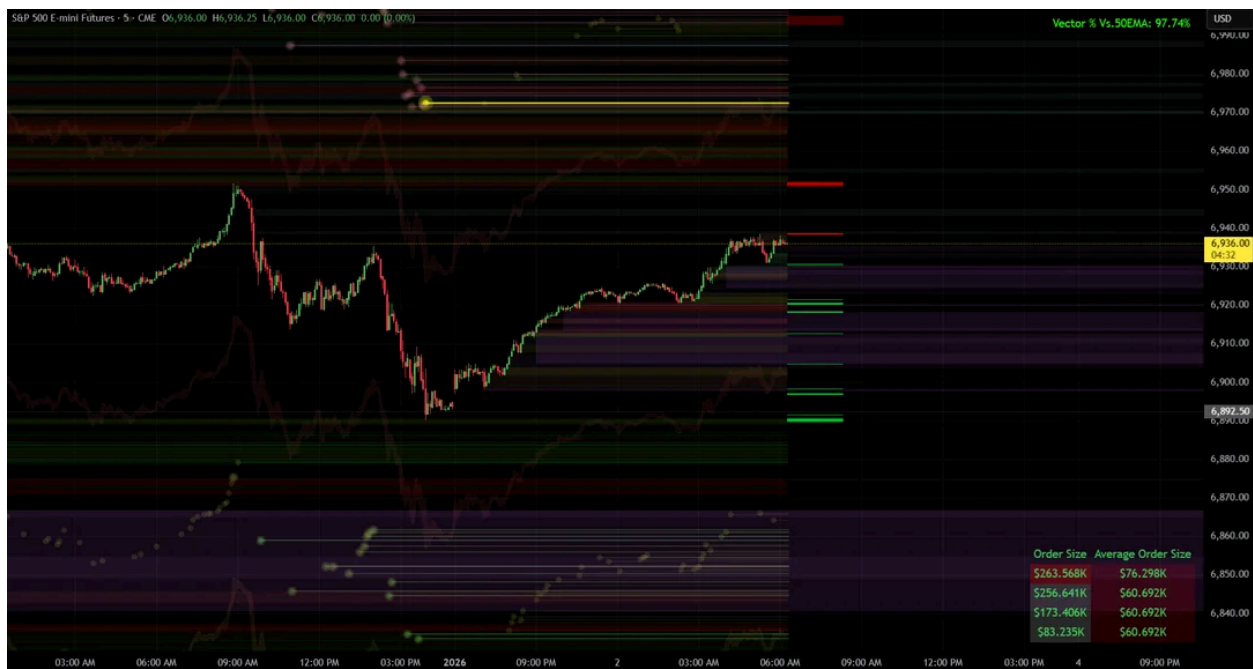
If you're a dev like myself, use that.

Also, comment down below if I should try my luck with all-encompassing post where I touch on as many things as possible. I feel like the algorithm will just wash it down or kill it entirely, so, what's the point of posting if it doesn't reach people.

I'm happy to answer questions about how I set up the heatmaps or what specific divergence settings I look for. Just drop a comment, I'll answer it.

(Obvioulsy NFA, just how I view the auction. Trading is risky (especially futures and options), don't blow your account))))

Post 3 months ago - Quit my mid dev job to trade full-time. Why I think patterns don't work for most, and how I engineered a semi-automated order flow system (Logic breakdown + code concepts).



Well, here comes the big, well-written and organized post. I'm writing this out mostly to clear my head and document the logic, but I hope the Reddit algorithm doesn't bury it immediately because it's too long. Anyways..

The Reality Check

I posted my P&L + strategy early last week, got a flood of comments and inbox. I'm seeing a massive amount of people here trying to trade based on "visual patterns" like wedges, flags, or whatever head and shoulders formation they see on a 5m chart. I did that for almost two years (prior to that, I was just trading vibes, which was actually better lol...). I lost enough money doing that to buy a nice car. I was literally donating money to the market makers every morning

The whole game turned 180 degrees when I realized that **the charts you are looking at are literally designed to trap you.**

My main advantage:

I'm a recently turned Senior full-stack dev. When I quit the corporate to do this fulltime, I treated trading like a backend engineering problem. The market isn't random lines, but a database of transactions. If you can read it (DOM, order flow), you can see where the big players are positioning

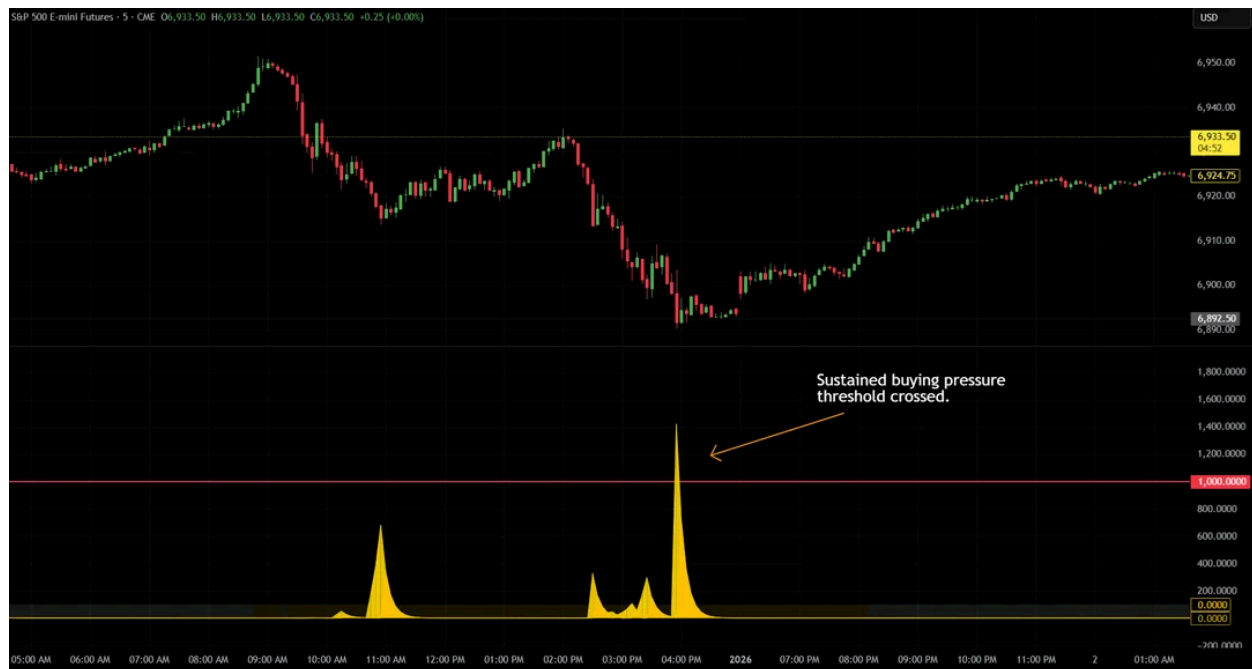
The system Logic

I wanted to share the logic behind the system I've been using for year and a half. It's an order flow sequencing model that mostly runs on futures (ES/NQ) but the logic applies to anything with liquidity

The thing is, I'm super lazy. I don't want to sit there staring at candles all day until my eyes bleed. I built a script in Pinescript (with some heavy lifting using arrays and lower timeframe request security) that runs in the background. It processes tick data and pings me when the volumetric pressure hits specific statistical thresholds. I just open my broker to execute.

And for the devs asking, yes, this is on Tradingview, but I had to jump through massive hoops to get the tick replay accuracy right

without it repainting. It's mimicking Rithmic data feeds I used in the past for backtesting and if you still use them, keep using them for now.



Why Support/Resistance is a Myth if it's not Somewhat based on order flow

Everyone draws the same lines. That's why you get stopped out by 2 ticks before the price reverses. Call that liquidity engineering

My script scans historical volume profiles to find high density nodes areas where institutions have previously transacted massive volume. These act as magnets. If price isn't interacting with one of these major liquidity zones or the session point of control (POC), the system doesn't look for a trade. I'm not trying to catch the middle of the move, I'm trying to catch the exhaustion at the edges where the retail traders are puking their positions.



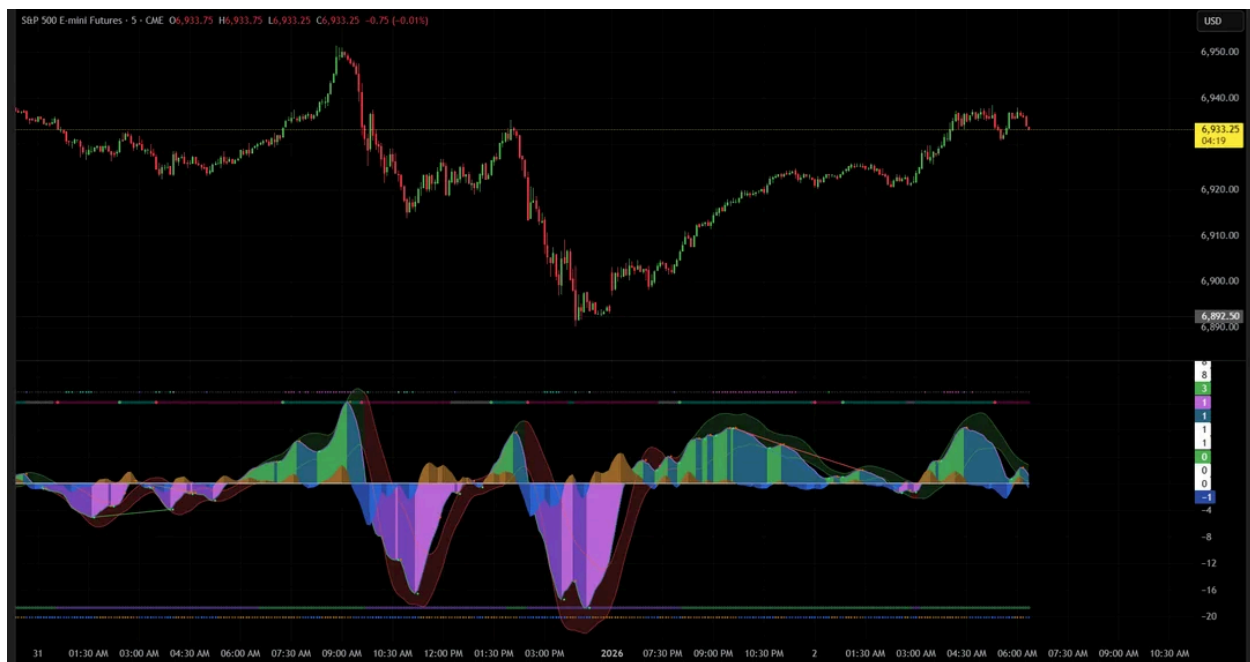
The Cheat code (Delta Divergence)

This is the hardest part to explain but the most crucial. This is why you lose trades that "looked perfect"

I don't just look at candle closes, but I track the sequence of tick trades within the candle coming through the time and sales. The script monitors the speed and size of incoming orders to calculate delta divergence.

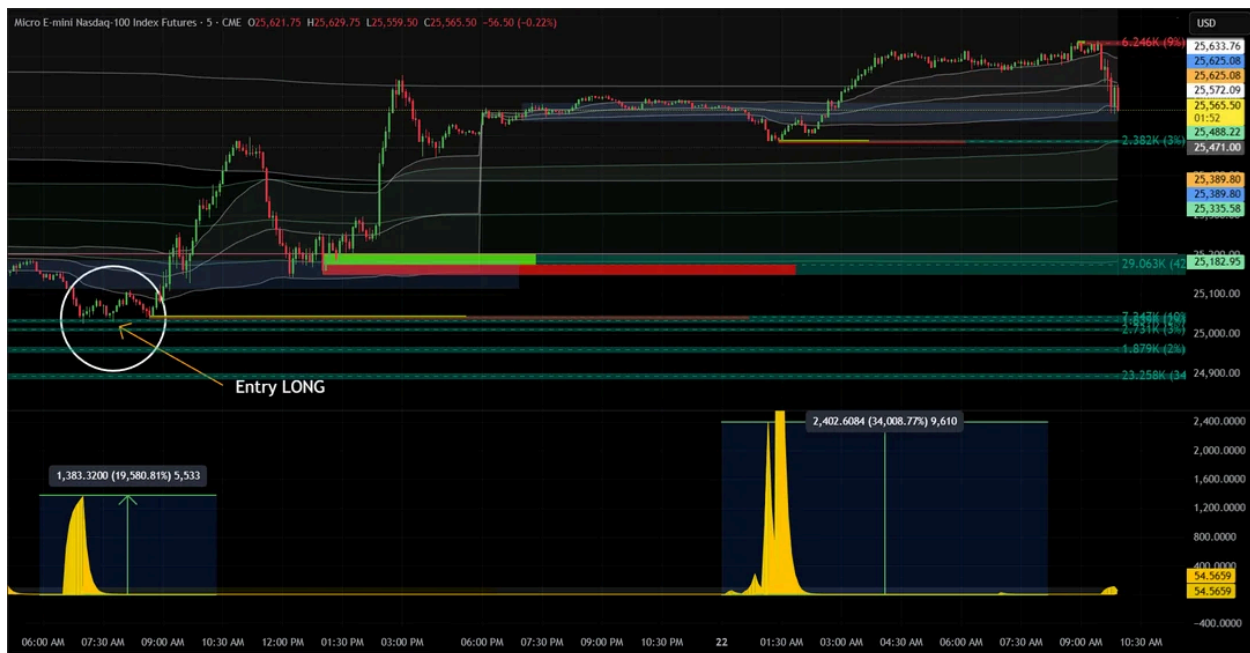
For the setup, imagine price is crashing HARD.

You're panicking and obviously, you sell. But the script sees that while price is making a lower low, the cumulative volume delta (aggressive selling) is flatlining or ticking up. It basically means sellers are dumping everything they have, but passive limit buyers are absorbing it all. It's a trap. It's all a trap. A trap. In reality, the sellers are exhausted



The human eye cannot process this data speed manually on NQ. By the time you spot a divergence on a standard footprint chart with your naked eye, the HFTs (High frequency trading algos) have already frontrun the move, taken all the liquidity. My script calculates the variance, checks the tick sequencing, and sends the final alert

I've been refining this logic while shadowing a few traders live on calls this year, and the biggest realization was that **the script acts as a filter for stupidity.** literally. Stupidity AKA emotions and "vibes", by the way. Essentially, If the math isn't there, I don't trade. It removes the emotion. I don't have to "guess" or "feel" the market



2025 stats (Since people asked)

- **Gross profit:** \$162,300
- **Net profit:** around \$127,000 (after data fees, commissions and tax set aside section 1256 contracts are a lifesaver for taxes, by the way, even if you have a job).
- **Win rate:** 55% (This sounds low to beginners, but my Risk:Reward is 1:2.5 minimum. I lose often, but I lose small and I win big. I should also note that my current win rate is 60-65% as I've refined the system in the year's interim)
- **Profit factor:** 2.53.
- **Max Drawdown:** 6.2% (I had 2 bad weeks last year, surprisingly in the chop and low IV - after this, I've implemented IV filter and Z-score)

Can you build this yourself?

If you are a senior dev with experience in financial data modeling and PineScript arrays, yes, you probably can. You need to understand how to map `request.security_lower_tf` arrays to capture intrabar volume without hitting the execution time limit on TV. This was

so much of a headache. If any of you guys are fixing to go this way, let me know, I'll help you with the arrays.

In addition, It took me about 8 months of iteration to stop the "fake" signals during chop.

A lot of people ask for the source code or the indicator setup. Look, I don't put this on public github for a reason. One, it requires a specific setup to work (it's not plug-and-play), and two, I protect my edge. If 50,000 people start front-running the exact same tick divergence signal, the alpha disappears.

This is why you can only have a few dozen, maybe a few hundred people trading the same thing. I keep my circle extremely tight, maybe overly tight.

And same thing happened with a lot of strategy-based frameworks here, like the initial iteration of ICT and SMC that worked, but flatlined after tens of thousands of people started entering at the same time, placing stops at the exact same level, basically providing liquidity for the hedgies and MMs

Anyway, hope this breakdown helps you understand why "buying the dip" works sometimes and kills you others. It's usually just order flow absorption.

Post Comments

Guys what OP is talking about is Cumulative Volume Delta (CVD), a simplified version is already available in tradingview just search for CVD. It's just a good approach but you must try to merge it with your own system to amplify the odds. It's not the Holy Grail!

This. I do a home-brew version of this. but i take the "stare until my eyes bleed" and after a few years created enough disciple to "just know" when the market is statistically in my favor.

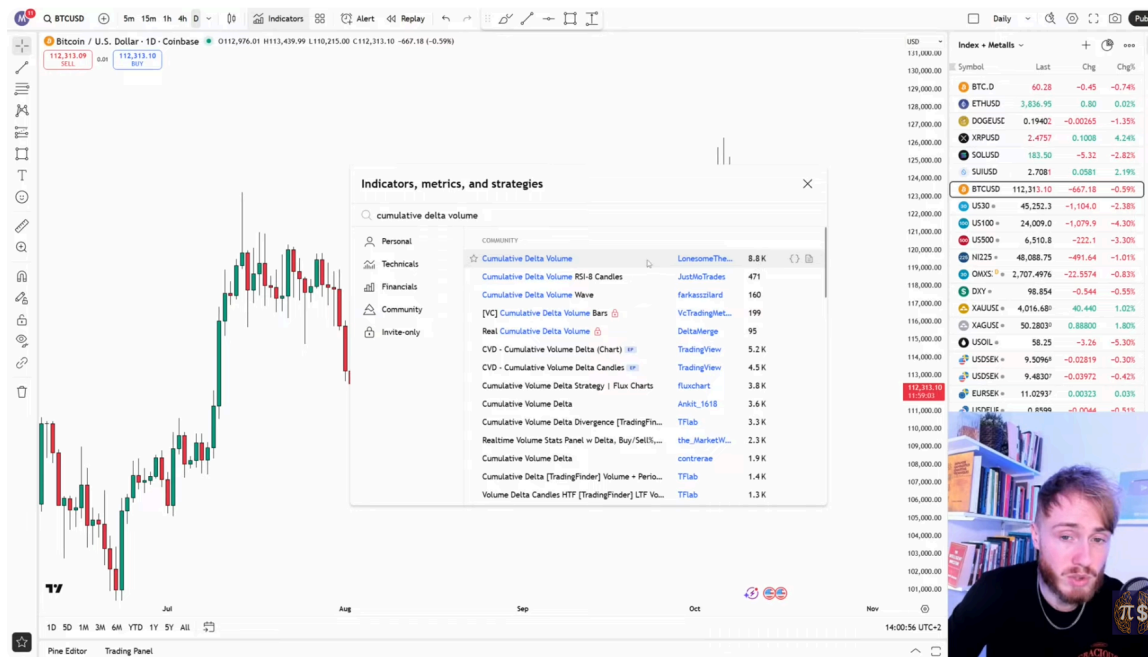
OP is for sure a more quantitative guy, and im more of a qualitative guy (AKA vibe trading). The speed, velocity and what spreads are doing matter *in-relation* to price action/levels. But also as OP states, the

market will eat your stop and reverse 2 ticks later. Levels *arent hard*, theyre a zone, a "vibe". You need to have some +/- room at these levels. OPs just using math more to show this. CVD helps visualize it. Another move is to have what you're trading on two charts, one at the time frame you like, and another is "tick based" charting. Once X amount of ticks are traded, it creates a candle. during time of volatility/speed, it quantities it. You can actually visualize the DOM better. I still heavily rely on just looking at a naked chart though.

As much as we like to put math and signals to our trading. Unless you're using algo's, you're trading to a "vibe" to a degree. Being up front and having enough experience to "know" the % are in your favor is the move.

The biggest edge you can have on the market, is time staring at the market.

I cant verbalize a lot of the "entries" i take. i just "know" because ive seen the market behavior at these levels, at this trading time (Open/Close/Globex open/London Open), with the fundamental backdrop (geopolitical news/political news/Earning season/Vix at this level. so i just "sort of know" what is the most likely price action to materialize.



Order Flow on TradingView with CVD [Buy & Sell Volume Exposed]

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[OP] - Agreed. it's as close to the holy grail as anything gets, but still not the holy grail without regime and bias filtering.

I am a senior dev as well with about 5years in manual trading. So I want to incorporate my dev skills which lead me creating a few indicators on TradingView. Nothing crazy as what you have shown.

My question which maybe off topic:

How does someone get started in this? Was this done with a particular tech stack or was everything done within TradingView?

If you've created some indicators on Tradingview already, that means you know the basics and are on the right path. You just need to formulate a strategy that works. Send me your email or something, I'll send you the code (heatmap or the momentum-based CVD only, though, can't do the main system), have a look at it to get inspiration on how to create something similar but the one that fits your strategy.

You've basically been trading for 5 years without using your main edge. Well, better late than never.

And this was mainly done on Tradingview, although for backtesting you need rithmic data for it to be tick-based. On TV, I hook data from CME data feed (costs \$6/month) and it's second-based, which is more than enough for my intraday system (it's not HFT to need tick-based everything).

I pulled about 3 years of clean tick data. The bottleneck isn't rithmic thouuhg, it's querying that much tick history without it lagging. Had to dump it into a DB to backtest it efficiently

From Trading vibes to being able to find an edge and to build all of this...it wasn't easy i think , how did you learned...Courses , books , videos

No courses, no books, no videos. I think I've answered your question before, something similar.

I'll say it again for the public record: If you have experience in development, use that.

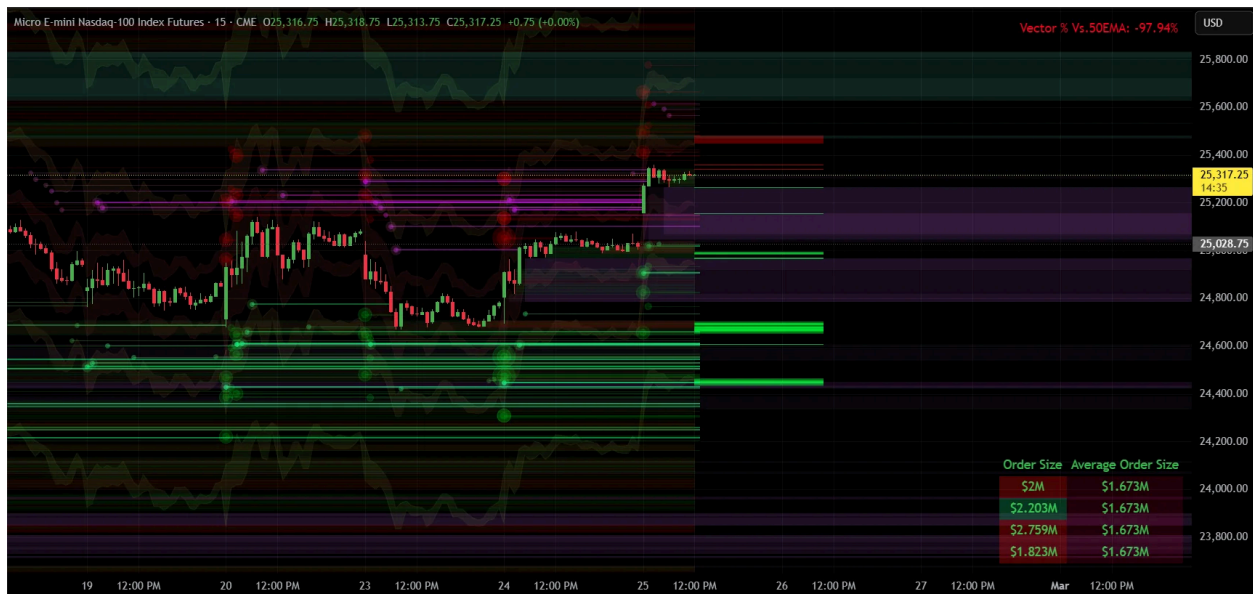
If you don't, research order flow, DOM, VWAP and start implementing it.

Try to automate as many things as possible, when it comes to strategy conditions detection.

USE MTF. non-negotiable.

These are a few must-do's off the top of my head. Again, I have only read a few dalton books (mostly before I became profitable - it is useful, but it's not straightforward), watched almost no videos and taken no courses.

Post 2 months ago - Why your support levels usually fail (and how to see the limit orders). Script included



If you keep getting stopped out by wicks, again, you're trading retail patterns against algorithms that hunt those levels because they know the retail liquidity is there.

I stopped trading patterns and started tracking resting liquidity. liquidity. I coded a custom Tradingview tool to visualize the order book directly on the chart.

The horizontal lines show passive liquidity (resting limit orders)

The bubbles show aggressive volume stepping in at those levels.

I keep my entry/execution logic private for my own circle, but this visualizer is free and will help you stop guessing where the big orders are sitting - liquidity, which is all trading's about.

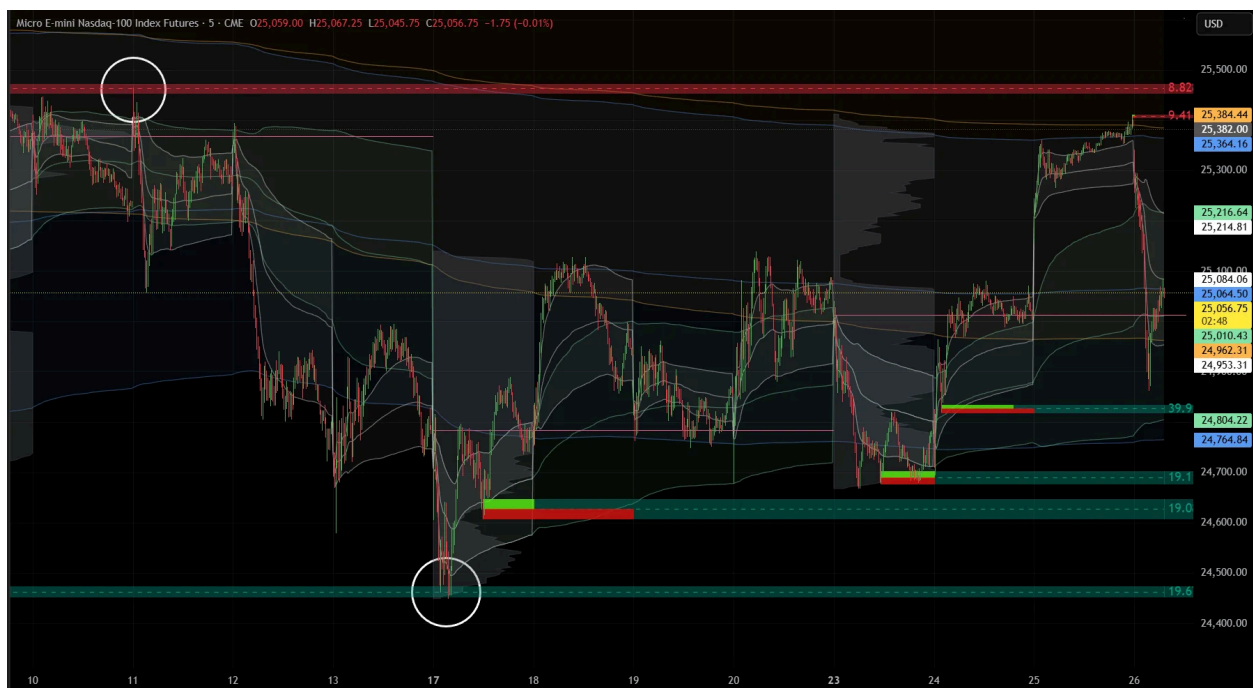
Note that we don't have tick data on TV and this simply estimates the DOM, and does it pretty damn well. Filtering logic included

Drop a comment and I'll send the pastebin link.

I've backtested dozens of trend strategies and I personally hated the drawdown and the wide stops. None of the strategies had the equity curve i was looking for (out of the few that seemed profitable in the long run), waiting for a trend to "confirm", usually means you are entering late

I trade mean reversion on NQ. I don't wanna use a wide stop. To get surgical precision, you need to see exactly where the heavy limit orders are resting and where the absorption is happening in realtime. a naked price chart can't show you the order book.

Post 2months ago Got tired of the emotional rollercoaster, so I coded my order flow edge into an automated system. This is the reality of algorithmic trading.



I've been trading ES and NQ for a while and like most people, my biggest leak was over leveraging, revenge trading, getting chopped up in the 1 minute tf noise.... psychology

My background is in software engineering (mid to senior dev) so about 2 years ago, I finally decided to remove myself from the equation entirely. Spent 3-4 months hard-coding my ideas, backtesting dozens of both well-known and

custom strategies, until I settled on the mean reversion strategy I've been using for the past 19-20 months (volumetric liquidity and order flow), coded it into a fully systematic script on Tradingview. While I primarily prioritize ES/NQ futures solely for the tax benefits and 23/5 trading, the edge applies to any liquid asset and I still use the system on high market cap stocks and index options when the volatility filters trigger (as shown in the statement image)

Systematizing your trading actually changes the reality of it in a few ways. For one, the win rate isn't what you think. Not even close lol. Everyone looks for 80% win rate holy grail, but my system hovers around 40-45%. Because the logic only triggers on severe delta divergence at key liquidity zones, the R:R is strictly 1:3+

Drawdowns are also boring now instead of terrifying. When you trade manually, a 3-trade losing streak feels like the end of the world and you almost certainly start tilting. When a system does it, you just look at the backtest data, confirm it's within the standard deviation of the equity curve, calm your horses.

You also actually get your time back. I no longer sit at the screens at 9:30 AM EST with high cortisol (had to use the reference...). The script filters the noise, identifies the volumetric exhaustion and prints the signal. I just execute, or let the API do it as I've automated the strategy, although only for the last few weeks.

If you're currently struggling with the mental side of trading, your best bet is to mechanicalize your rules. Take your entry criteria, write them down objectively, and backtest them blindly. If you can't code it, you probably don't have a real strategy, you just have a feeling.

Happy to talk shop, answer questions about volumetric logic or share how I structured the delta calculations if any other devs or traders are trying to build out their own systems

Q and A

May I ask some questions about: max losing streak, total capital, max drawdown, and how you pull the liquidity data level (which data feed, tool, etc). Thank u

Historically the max losing streak is around 7 or 8 trades. Sounds rough but with 1:3+ R:R, one good mean reversion catch makes up for it pretty quick. Max drawdown hovers around 20-25% in backtesting (5 year backtest) and around 12% in my live trading experience. I started with \$2k just to afford micro

contracts, currently I run a personal account that's currently at high 5 figures, should make it to 6 end of this month, also got into prop firms and currently have 2 funded accounts, one's \$50k and the other's \$150k. Though bottom line is, it scales to whatever as long as you keep risk at 1-2% per setup.

For the data and tools, everything is built natively in Pinescript on Tradingview (It's a house framework entirely based on backend stack I'm great at). I just use the standard real-time CME data feed through TV (costs \$8/month). The script basically calculates the volume delta and maps the liquidity zones directly on the chart

Let me know if you want me to break down how the delta calculations are structured under the hood, always down to talk shop, especially if you're a dev but not a mandatory obviously.

The frequency of trades, like a few trades everyday or the system is selective?

Thank u for your detail reply. I appreciate much

Averages out to about 0.8/day. Since one of the triggers of the logic is severe delta divergence at those specific zones, there are times the system sits flat for a whole week without a single trade. Also when the market's making all time highs, the system doesn't print any signals due to some other specific conditions of the logic I can't discuss, so that also cuts the frequency when we're just going up without any corrections. Sometimes that goes for over a week

I just spun up a custom ultra lightweight node server to catch the Tradingview JSON payloads. It parses the ticker and trade direction, then routes it directly through the IBKR REST API.

If you're using prop firms, Topstep for instance has licensed third parties that bypass its TOS restrictions, like Traderspost or Pickmytrade, meaning you can use those too. I still haven't automated anything but my personal account, so how it works on prop firms is still something I'm researching, but the bottom line is that you can definitely use third parties to automate on prop firms, instead of using VPS (which usually goes against TOS of most prop firms).

Your Script gives you Orderflow data level 2 right ?

Nope, not full level 2. TV Pine doesn't expose the DOM like that. It calculates the order flow using aggressive tick volume (up/down delta) at key price levels.

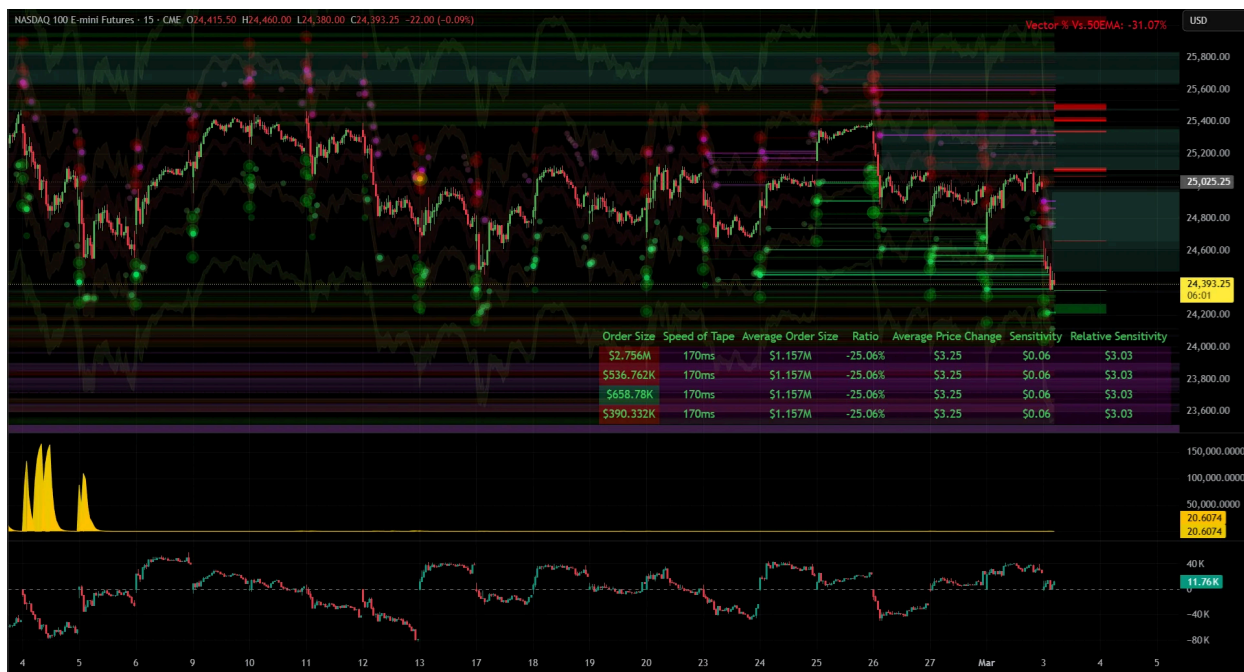
It's a mean reversion model, not an HFT latency bot / arbitrage, so standard realtime tick data is plenty for the volumetric exhaustion signals I look for on daily basis

Hey thanks for sharing. You are options buying or selling? Delta divergence, momentum and volume is the key logic I am currently working on. Any tips to cut losses?

Mostly trading the futures contracts directly, but when the volatility filter triggers for index options, it's strictly buying. Selling premium doesn't really fit the 1:3 risk/reward structure I rely on.

As for cutting losses with that specific logic, the best tip I can give is tying your stop strictly to the invalidation of the liquidity zone, not a random dollar amount. Last (lookback of 20 candles, per se) swing point is also surprisingly decent stop. But my recommendation is still to set a hard mechanical stop right outside the order block and let it take you out if the volume doesn't actually exhaust. Don't step in front of the train if the momentum is still heavily one-sided

Post 2months ago NQ weekend gap + what the system flagged



Sunday open was headline volatility. Thin liquidity + the Iran situation = exaggerated move.

Once liquidity normalized, a few things showed up in the volumetric layrr.

1. HEAVY sells into 24,650–24,600... no displacement.

5m absorption alerts started firing there.

Delta expanded aggressively negative, but price stalled instead of accelerating.

When the model sees aggressive market sells without range expansion, it tags it as passive absorption.

That zone lines up with lower boundary of recent value, repeated demand late Feb, edge of the broader 1,500pt range. If sellers had real control, we wouldn't be compressing here, that's for sure.

2. 24,650 is the inflection

Below it continuation logic opens toward 24,350 & 24,240.

Above it and holding rotation probabilities increase

Nothing predictive however, just conditional structure.

3. 25k still the macro pivot because friday never got acceptance below it.

If 24,800-25k gets reclaimed with volume, this likely becomes a liquidity event.

If we build value under 24,650, that's the first legitimate structural shift since early to mid Feb I believe

What the dashboard is tracking is:

Sell aggression vs displacement

VWAP reclaim efficiency

Liquidity stacking (essentially above vs below)

Speed of tape + order size divergence

In general, gap environments tend to produce sharp two-way trade. The first impulse is usually not the direction

Though am I talking in the mirror? I mean, I'm curious who in this sub is running CVD / absorption models vs just candles.

Post (2 months ago) I spent 2 years bleeding money on "price action" until I coded a script to track the actual order flow (ES/NQ strategy breakdown)

Again and again, I constantly see traders in here trying to pass prop firm evals or grow small accounts by trading wedges, flags or drawing diagonal lines on a 1min chart. I did that for my first two years and lost quite a sum, to put it mmildly.

My background is in software engineering, and I only turned the corner when I stopped trying to predict candlestick patterns and started treating the market strictly as a data problem. Again, maybe it's just me, but I wanted to break down the mechanics of the automated system I eventually built. It runs exclusively on ES/NQ futures. The most important part is that I don't sit at my desk staring at DOM and order flow. The script runs the calculations in the background and only alerts me when the statistical variance is heavily skewed in my favor.

The baseline stats (Trailing 12 Months):

Win rate: 48% (If you are looking for an 80% winrate, you're going to be disappointed. The entire edge is in the asymmetric risk:reward). Profit Factor: 2.52
Average RR: 1:2.5 minimum per setup. Max drawdown: -5.8% (Hit a rough one last summer in a chop (the chop was so bad that even my mean reversion strategy took a small loss)).

The Core Logic: I completely abandoned traditional "support and resistance" because those are just lines. Instead, I built the script to track aggregated volumetric liquidity, which basically plots where the massive institutional limit orders are actually resting.

If price isn't pushing into one of these heavy liquidity nodes (and into the session VWAP bands), the script stays flat. There is no trade.

When we do reach a zone, the script calculates the cumulative volume delta (CVD) at the tick level. The human eye is way too slow and imprecise for this because by the time you see a reversal candle close, the algos have already taken it. The script looks for severe divergence + the price falls into one of the filtered order blocks + VWAP HVN/LVNs. For instance: Price pushes to a new low into a liquidity band, but the aggressive sell volume (CVD) is actually making a higher low. That means the aggressive market sellers are exhausted and just slamming into a massive wall of passive buy limit orders. It's pure absorption. Once that absorption is mathematically confirmed, the script prints the signal. Stop loss goes strictly a few ticks below the absorption wick. Target is the session mean (VWAP).

The tax advantage (Why futures): Whenever I talk about this, people ask why I don't just run this on SPY options. So to just get that out of the way, the answer is IRS Section 1256 AND simplicity.

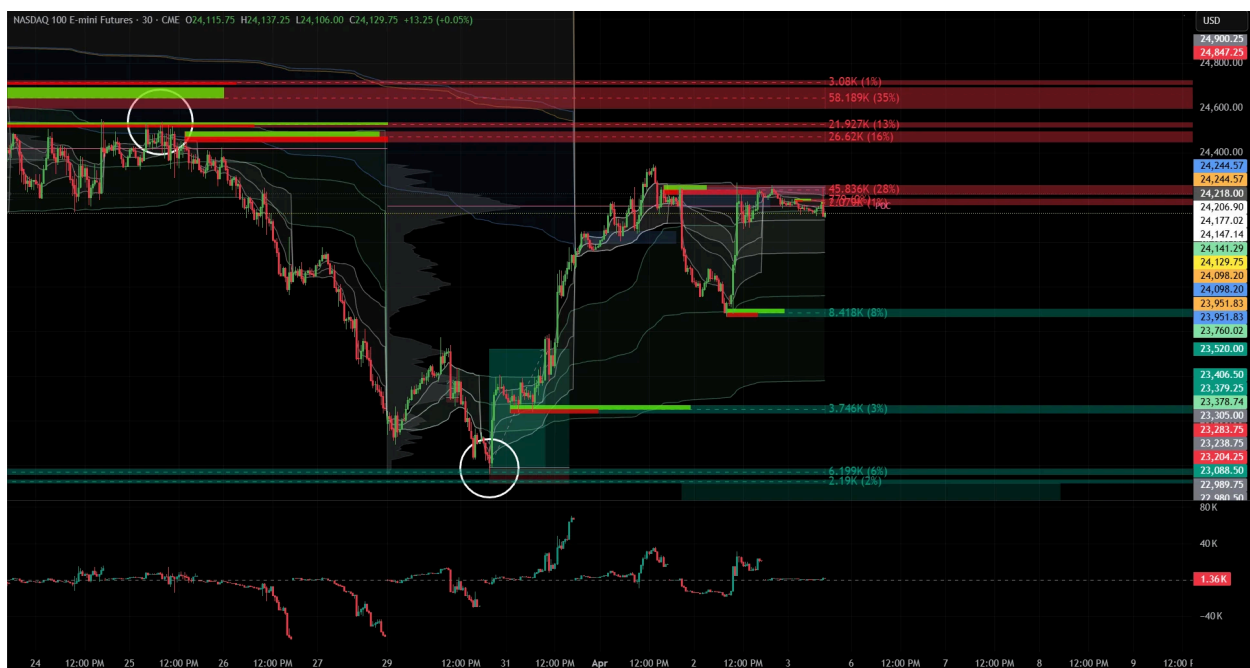
Trading futures means 60% of the profits are taxed at the long-term capital gains rate, regardless of how long I held the trade. Living in a state with income tax, this saves me over ten thousand a year compared to scalping stock options. Plus yeah, I don't have to deal with the wash-sale rule, theta decay, or IV crush. I just trade the raw data. So, you can get the same tax treatment on SPX rather than SPY, but then you still have to deal with all those options-specific pricing mechanics (like the Greeks) that the algo doesn't account for.

I essentially treat this as a high-paying, remote engineering job that scales more than anything I know. It took a long time to build the infrastructure, but the hardest

part was just learning to sit on my hands, cut out emotions and simply trust the code when it told me to stay out of the market.

Happy to talk shop or answer questions in the comments for anyone trying to make the jump from drawing patterns to tracking order flow.

Post (1 month ago) - Order flow breakdown and entries on recent Futures volatility (pinpointing the Monday reversal)



Market was heavy last few days, but NQ/MNQ offered a textbook mean reversion setup if you were watching the tape instead of lagging indicators.

Wanted to showw how the liquidity heatmaps flagged the rotations

The top (first circle) shows exhaustion at the 24400 supply zone. Heavy mostly passive selling absorbed the aggressive buy pressure, creating a massive (40+ candles lookback) delta divergence. Once bids pulled, we went down through the low volume nodes (LVNs). We didn't enter here, other conditions weren't aligned.

The bottom (second circle): Usually this is the place where retail gets chopped. The good old "Trying to catch falling knives". Order flow showed that the price

drove into a thick liquidity pool at 23088 (the green order block, 6% density).

The system flagged this not just because "it hit a line", but because other conditions also aligned. specifically, aggressive market selling met passive absorption at that level, that's also visible on the CVD bottom of the chart, though the one the system accounts for is MTF. That coupled with a spike outside VWAP and a delta divergence, the created imbalance forced a very swift snapback

If you're a dev and been trading a few years to know market structure on a relatively high level, build these tools. Automate entry condition detection. I built these tools on Tradingview in their native Pinescript to automate this order flow logic because standard TA was obsolete in my case at least. Before someone asks, no, this is not HFT bot, therefore we don't need level 2 order flow data. It is completely irrelevant in this case. I'll be in the comments.

Q and A

the tools you coded, what are those? trading view indicators or something else?

Yup, it's on Tradingview, though not in the indicator/study() format. It's in strategy() format to execute on the alerts it sends.

so you do automated trading? A bot is a executing trades for you?

For the first year and a half, I was just using the system for alerts. The script would ping me when the volume data and structure aligned, and I'd physically click buy/sell myself. I actually preferred doing it that way for a long time just to keep a pulse on the execution and be in front of the charts for a longer period of time. It's only been the last few months that I finally hooked it up via API to fully automate the entries and exits, and I still keep some of the execution (just clicking buy/sell) manual

Any advice for someone with a systematic back tested way of finding good entries but not sure what to settle on for a exit strategy

If your entries are systematic, then I say don't use arbitrary dollar amounts.

You can target the session VWAP. If you enter at the 2nd standard deviation band, then the mean is your default exit.

You can also use volume profile, I'd specifically use the multi-timeframe volume profile and set my limit order the next major (or take out the word major, you want to be as precise as possible) HVN.

You can also hold the position until you see CVD diverge against you (e.g your long position makes a higher high but the buy volume makes a lower high), though this would probably work in tandem with one of the 2 listed above.

If you want my advice, I'd pick the one that keeps your RR at least 1:2 (at least) and backtest it thoroughly. Good luck

Are you still using the strategy where if the price moves away too many standard deviation then you do mean diversion simply put

Yeah, same system, but that's just the prerequisite. Fading a standard deviation band by itself will blow your account on a trend day.

The SD extension just puts the script on high alert for a reversal. It runs 4 (or 5, depending on how you put it) other volumetric and order flow calculations on the backend before it confirms a signal. You can't really "simply put" a systematic edge like that.

(interesting comment, no reply from OP)

Hey, I was fascinated by your algo journey and I tried to re create the algorithm. Can you please take a look and let me know what you think ?

Core idea: Mean reversion — when price gets stretched too far from the average AND big sellers/buyers are exhausted, fade the move back to the mean. Entry conditions (all must align within a 30-minute window):

Regime check: ADX must be below 25, meaning the market isn't in a strong trend. We only want to mean-revert in choppy or mildly trending conditions. Liquidity zone: Price must be sitting in a high-volume node — an area where institutions have historically transacted heavy volume. Built from a 60-bar volume profile with 50 price bins. If the current bin's volume is 1.6x above the average bin, it qualifies. VWAP extreme: Price must be at or beyond the 1.75 standard deviation band of the session VWAP. This means price is statistically stretched from the day's average. CVD divergence: Price is making new lows (or highs), but the cumulative volume delta disagrees. For a long: price hits a new low in the last 14 bars, but CVD is actually higher than it was 14 bars ago — meaning aggressive sellers are dumping but passive buyers are absorbing it

all. Calculated from 1-minute intrabar data. Momentum confirmation: A smoothed rate-of-change oscillator must be crossing above its signal line OR be above signal and rising — confirming the reversal is starting.

Exits:

Target: Session VWAP (the mean). If VWAP is too close (less than 1:1 vs stop), falls back to a 2.5x R:R fixed target. Stop: 1.5x ATR below entry End of day: All positions closed at session end

Risk management:

15-minute minimum cooldown between trades ATR-based stops that adapt to current volatility No trading in strong trends (ADX filter)

Backtest results (MNQ, 1 year):

117 trades (~1 every 2 days) 39% win rate 1.25 profit factor +\$1,872 on 1 MNQ contract Max drawdown: \$1,870

Post (8 days ago) Here's how being a dev helped me stop bleeding money on feel and systemize my edge.

There's still A TON of people on here trying to trade based on "feel" or drawing arbitrary, patterns. I was doing exactly that a couple years ago and blew a decent chunk of change for me at that time. Again and again, I say, things haven't changed slightly until I started trading statistical variance instead of "feel" and "discretionary".

Just wanted to lay out and update the framework of the system I've been running. Basically, it's a mean reversion setup built for Futures (ES/NQ) that targets overextended liquidity zones.

One major thing though: I'm not gglued. I have a script running in the background that does the heavy lifting, and it only pings me with an alert when all the conditions line up. Then I step in and execute.

Win rate is sitting around 41% (the real edge is the R:R, not the win rate, but my usual win rate is usually around 50%, just a bad streak so far)

Profit factor is 2.40

Max drawdown hit -6.8% at its worst.

The core logic:

I ignore standard "support and resistance" lines. Not trashing them, they are still based on liquidity, but the math clearly prefers something more straightforward, **like volumetric zones**. The baseline assumption here is that **price just can't hold a move outside 2 standard deviations (2SD) of the session VWAP unless there's aggressive market order initiation pushing it**.

If that aggression dies off, a reversion to the mean becomes very highly probable. Though just this without coupling it with other conditions below will get you high win rate (if you're trading 1:1 RR), but the few losses will sting more.

This is why I don't take a trade unless the script confirms ALL conditions are met, meaning no manual hunting.

For the statistical stretch, price has to break outside the 2SD of the anchored VWAP. That puts us in outlier territory.

This extension needs to poke into a MTF (usually 30m or 1H) low volume node (LVN) or an older high volume node (HVN). I need proof there's a liquidity backstop waiting.

As price pushes to a new extreme outside the bands, the cumulative volume delta (CVD) has to show a divergence with 30-50 lookback no less. E.g. price drops to a new low, but CVD forms a higher low. That shows passive absorption limit buyers stepping in to eat the aggressive selling.

Another condition is a reversal off a volumetric buy/sell side filtered order block. I can't give away too much of the sauce on this, but let's just say, the aforementioned order block should NOT be freshly formed.

On top of this, a must: This year I started incorporating the Hurst Exponent. It significantly bumped my risk:reward ratio while maintaining around the same PF.

For the trigger, the script waits for a 5m candle to close back inside the geometric range of that liquidity zone.

Then pings me and I enter right away.

Entry is a market order the second that reversal candle closes.

Stop loss is a hard stop right behind the absorption wick. If the wick breaks, the absorption was fake, the script cuts it instantly.

Take profit targets the session VWAP first, then the opposite 1st std band. I've also been testing simple RR-based TP (SL-anchored, of course).

Advice for devs:

DO THIS. If you're a mid or senior dev with basic backend stack, you can likely build this kind of a system in Tradingview Pinescript within a year. Takes hard work, but the results will be worth it.

You could theoretically do this manually, but tracking live delta divergence, 2SD bands, heatmaps, order blocks, hurst and z-scores is, unfortunately, simply impossible.

That's why a script is your gatekeeper. If the math's off, no alert. I could fully automate the execution, but I honestly like the peace of mind of hitting "Buy" myself just in case a COVID or 9/11-like black swan event hits.

note on taxes:

A lot of folks ask why I avoid options. First off, why I avoid spy is IRS section 1256.

Because I'm on Futures, 60% of the gains get hit with the lower long-term capital gains tax rate. Also, no washsale rule means I can scalp a level repeatedly without the accounting headache you get with SPY options.

I could trade SPX, but then I have to account theta, time decay, and premiums. No thanks, I wanna get paid when I catch the right direction.

Some devs may laugh at this and may think overall take-home will be less than some salaries in the field... no. If you don't repeat the same mistake I made by using the profits to pay off my monthly expenses and instead reinvest it and let the compound interest do its thing, you'll eventually do better than any dev on the face of this world.

Systemize your trading.

Edit: I'll be back at my desk in a bit, if anyone wants to see what it looks like on the chart to visualize the setup, lmk, I can probably grab a screenshot.

Post (1 day ago) I quit my dev job to trade full-time. 6 months later, here's the data and the order flow model I use (Performance Update)

5/5 months green

Win rate: 45.5% (though as the April screenshot shows, it can dip while staying profitable)

Avg RR: 1:3.9

Profit factor: 2.1

The core of my system is still a mean reversion model on Futures (ES/NQ, mostly NQ).

I don't use standard technical analysis. My system is based entirely on order flow. I coded a custom suite that acts as a gatekeeper. It waits for the price to hit statistical volume extremes, combined with delta divergences and order blocks. The entry window is extremely small, allowing for a very tight stop loss and a massive RR ratio.

If the script doesn't print a signal based on delta and order flow data, I sit on my hands and don't trade. That's really all it takes.

If the script doesn't print a signal based on delta and order flow data, I sit on my hands and don't trade. That's really all it takes.

I've made minor changes to the system since then, but the core logic and entry conditions have stayed the same.

I still rely heavily on the 1256 tax benefits that come with futures (60% long term and 40% short term capital gains split) and ignoring the wash sale rule. I'm based in Richmond VA, my state tax is around 5.75%.

I did make a big shift recently. So about 3 months ago, I started trading stock options too, to increase my trade frequency. Yes, I know I just praised the tax advantages of futures lol. But the market has been presenting clean setups outside of my futures zones, so, extra profit outweighs the tax disadvantages.

The compound interest from starting with my initial capital has been crazy too. Started with a \$2k-\$3k. The hardest thing to do was cut off emotions and sit on my hands when the system said so.

If you're still trying to find consistency, my advice is, stop trying to predict the future with trendlines and start trading liquidity, order flow, volume... And if you're a dev, even mid or mid-to-senior, try building a system like this, I built mine in under 2 years of R&D.

Post I quit my dev job to trade full-time. 6 months later, here's the data and the order flow model I use (Performance Update) - Q and A

Can it be automated?

It's 100% rules-based, so it can definitely be automated. I currently execute manually to stay engaged with the price action, the system and also, if a black swan event were to occur (big ones, like 9/11), I can stop before it does automatically at 50% capital loss.

As for additional rules, nothing.

Trades per week

Used to be 4-5 trades a trading week on futures, it's now down to 1.5, maybe 2 trades a trading week given that the market's been renewing all-time highs every other week past 2 months. I'm sure the market will pull back for a month or two at some point, but, in the meantime, I've somewhat increased my trade frequency by applying the system to stock options now too, mostly highly liquid stocks, top 20 in terms of market cap.

I really appreciate this post! The world of trading is huge and I was having trouble deciding where to start. I downloaded one day of ES MBP-10 from DataBento along with 1-min OHLCV, will start looking for some patterns.

Q for you, if you're inclined to answer: are you using a platform with custom rules you coded, or did you subscribe to a data feed that provides live MBO / MBP and you crunch all the numbers on your PC in real time?

Thanks!

Appreciate the kind words, but I must answer your Q directly: the system is on Tradingview. I've backtested "lite" version of my script on sierra charts about 2 years back, before I'd finally start live trading it. Sierra to get the best data and all-around backtesting for years of lookback (Ninjatrader's also decent), but I still execute on Tradingview.

Few things I feel we need to get out of the way because you may be overcomplicating day trading and slightly confusing it with HFT (high frequency trading):

You don't even need tick data. Second-based data is more than enough. OHLCV is a great data point, but it's mostly used for trend-based systems. If you're building a mean reversion strat, cut that.

Your Q was very important, because building this by crunching all the running numbers on a PC would be an absolute nightmare and I don't know what your stack is, but I for one would NOT be able to do that with my fairly decent backend dev stack.

The data you have on Tradingview is laughable for HFT but completely sufficient for building a day-trading albeit profitable system, that's rooted in order flow, trend-based metrics, reversal metrics, etc... In my case, order flow.

What are you using for your data source?

I use Tradingview Pine Script (Tradingview) for the live execution logic because I want server-side alerts without hosting my own listener 24/7. Otherwise, I've used python (Pandas + NumPy) for data research. Sierra Charts also has great data but only used it once. Anyways, I exported historical data and ran the volatility models there first to find the optimal standard deviation settings. Although this has become less necessary as Tradingview rolled out some new subs last year that support backtesting of over 100k bars I think. But this also requires hard coding entries, which may take time for some scripts.

Anyways, I then took the backtesting data and logic onto TV Pinescript... I realized early on that trying to build my own charting UI in vanilla js or React would just amount to procrastination. Tradingview's infrastructure is unbeatable, so I just use their alert system as my backend trigger.

If you're building a day-trading system, not HFT, and if I were to give you an advice, I'd say, fully rely on Tradingview. For futures, they have \$8/month CME feed you can get. And Premium sub for smoother data, though I strongly believe that for a day-trading algorithm, this isn't fully necessary if your edge is robust-enough. You also don't need level 2 order flow data unless, again, you're doing HFT.

What would you recommend as the best way to learn how to do this?

If you're a dev, you can lean into order flow. Obviously, order flow is what I recommend because that's what I know the best, know it works and that's what has been working for me for over year and a half, consistently. Specifically CVD divergences, order block filtering, filtering liquidity fakeouts, VWAP... The most instrumental ones off the top of my head.

If you don't have a background in development, well, the only way is to learn it. A junior whose entire stack is wired around trading is infinitely more capable than a mid or even senior dev who knows nothing about trading concepts. Though however you approach this, it'll take time. At least a few years to get everything done, starting from learning code to coding a system like this, backtesting it and achieving profitability.

P.S As a very careful sidenote, I'd say, AI is developing so fast that it's very possible for it to be super useful for this endeavor in a year or two, BUT, currently, AI struggles heavily in house scripts used in trading, like Pinescript, MQL, Ninja, etc... Largely because they're either house scripts (so their syntax isn't mainstream), or they have libraries that AI cannot access. This is why currently, it seems impossible to build a system of this complexity using just AI. Maybe in a year, 2 years... it could be possible. I definitely see it.

What do you do when you are waiting for a trade setup?

Nothing. I don't even look at charts that much when the market's open, because I know myself, I know I may be tempted to take a trade before getting an alert from the system. So yeah, I just wait for an alert from the script and simply step in to execute. That's it.

I've been building something for myself the last couple of months, curious to know which data source you're using, especially having your system based in order flow and volume. Congrats on the success!

Thanks. For backtesting, I've used Pandas + NumPy and Sierra, Sierra has the best data hands down, but Ninja's more affordable option for 99% of the same results.

For execution, I'm using the good old Tradingview Pinescript. I know they don't support incoming APIs, or level 2 order flow, or proper tick data like say Ninja does, but Tradingview has the best interface and it's not even close. + You don't have to spin up your own listener.

Anyways, on Tradingview, you can subscribe to \$8/month CME data sub and if you want smoother data, you can get tradingview's premium subscription. That's all, but if you aren't day-trading and instead are doing HFT or something along the line, don't take this advice.

Yes, I personally recommend sticking with semi-auto. In my case, that is getting the buy/sell alert and me just stepping in to execute (buy/sell on brokerage). The script also sends auto stop loss via the alert, and take profit is just based on RR from that stop loss.

Tradingview has ultra simple alert functions built into it. That is the 'alert' function that's useful in a local scope, and the 'alertcondition' function in global scope.